

Tutorial -4

Q1: Consider the CFG

$$\begin{aligned} E &\rightarrow E + T \mid T \\ T &\rightarrow T \times F \mid F \\ F &\rightarrow (E) \mid a \end{aligned}$$

Give parse trees and derivations for each string.

1. a
2. a+a
3. a+a+a
4. ((a))

Q2: Use the languages $A = \{a^m b^n c^n \mid m, n > 0\}$ and $B = \{a^n b^n c^m \mid m, n > 0\}$ to show that the class of context-free languages is not closed under intersection.

Q3:

Give context-free grammars that generate the following languages. In all parts the alphabet is $\{0,1\}$.

- $\{w \mid w \text{ contains at least three 1s}\}$
- $\{w \mid w \text{ starts and ends with the same symbol}\}$
- $\{w \mid \text{the length of } w \text{ is odd}\}$
- $\{w \mid \text{the length of } w \text{ is odd and its middle symbol is a 0}\}$
- $\{w \mid w = w^R, \text{ that is, } w \text{ is a palindrome}\}$
- The empty set

Q4: Give descriptions and state diagrams of pushdown automata for the languages of Q3

Q5: Give context-free grammars generating the following languages.

- The set of strings over the alphabet $\{a,b\}$ with more a's than b's
- The complement of the language $\{a^n b^n \mid n > 0\}$
- $\{W\#x \mid W^R \text{ is a substring of } x \text{ for } w, x \text{ in } \{0,1\}^*\}$
- $\{X_1\#X_2\# \dots \#X_k \mid k \geq 1, \text{ each } x_i \text{ in } \{a, b\}^* \text{ and for some } i \text{ and } j, x_i = x_j^R\}$

Q6: Draw the PDAs for the Grammars of Q5

Q7: Give a context-free grammar that generates the language $A = \{a^i b^j c^k \mid i = j \text{ or } j = k \text{ where } i, j, k > 0\}$. Is the grammar ambiguous? Why or why not?

Give the description of a pushdown automaton that recognizes the language A .

Q8: Convert the following CFG into an equivalent CFG in Chomsky normal form

$$\begin{aligned} A &\rightarrow BAB \mid B \mid \epsilon \\ B &\rightarrow 00 \mid \epsilon \end{aligned}$$

Q9: Let $G = (V, \Sigma, R, S)$ be the following grammar. $V = \{S, T, U\}$; $\Sigma = \{0, \#\}$ and R is the set of Rules

$$\begin{aligned} S &\rightarrow TT \mid U \\ T &\rightarrow 0T \mid T0 \mid \# \\ U &\rightarrow 0U00 \mid \# \end{aligned}$$

Describe $L(G)$ in English and show that it is not regular

Q10:

Give a counterexample to show that the following construction fails to prove that the class of context-free languages is closed under star. Let A be a CFL that is generated by the CFG $G = (V, \Sigma, R, S)$. Add the new rule $S \rightarrow SS$ and call the resulting grammar G' . This grammar is supposed to generate A^*